

Securitize DSToken Rebasing Audit Summary

Prepared by Cyfrin

Version 2.1

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Securitize DSToken Rebasing. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated October 10, 2025, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the Securitize DSToken Rebasing code provided by Securitize. The findings consist of 1 Critical, 9 Medium & 23 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Securitize DSToken Rebasing
Repository	dstoken
Commit	f288c359ee85...
Fix Commit	f20fd8d0fc3f...
Audit Timeline	Sep 1st - Sep 19th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	0	0	0
Medium Risk	9	5	4
Low Risk	23	17	6
Informational	25	15	10
Gas Optimizations	20	11	9
Total	78	49	29

6 Audit Scope

The scope of this audit was limited to:

```
contracts/bulk/BulkOperator.sol
contracts/compliance/ComplianceConfigurationService.sol
contracts/compliance/ComplianceService.sol
contracts/compliance/ComplianceServiceNotRegulated.sol
contracts/compliance/ComplianceServiceRegulated.sol
contracts/compliance/ComplianceServiceWhitelisted.sol
contracts/compliance/IDSComplianceConfigurationService.sol
contracts/compliance/IDSComplianceService.sol
contracts/compliance/IDSLockManager.sol
contracts/compliance/IDSWalletManager.sol
contracts/compliance/InvestorLockManager.sol
contracts/compliance/InvestorLockManagerBase.sol
contracts/compliance/LockManager.sol
contracts/compliance/WalletManager.sol
contracts/data-stores/BaseLockManagerDataStore.sol
contracts/data-stores/ComplianceConfigurationDataStore.sol
contracts/data-stores/ComplianceServiceDataStore.sol
contracts/data-stores/InvestorLockManagerDataStore.sol
contracts/data-stores/LockManagerDataStore.sol
```

```
contracts/data-stores/RegistryServiceDataStore.sol
contracts/data-stores/TokenDataStore.sol
contracts/data-stores/ServiceConsumerDataStore.sol
contracts/data-stores/TrustServiceDataStore.sol
contracts/data-stores/WalletManagerDataStore.sol
contracts/issuance/IDSTokenIssuer.sol
contracts/issuance/TokenIssuer.sol
contracts/multicall/IssuerMulticall.sol
contracts/multicall/MulticallProxy.sol
contracts/rebasing/RebasingLibrary.sol
contracts/rebasing/SecuritizeRebasingProvider.sol
contracts/registry/IDSRegistryService.sol
contracts/registry/IDSWalletRegistrar.sol
contracts/registry/RegistryService.sol
contracts/registry/WalletRegistrar.sol
contracts/service/IDSServiceConsumer.sol
contracts/service/ServiceConsumer.sol
contracts/swap/BaseSecuritizeSwap.sol
contracts/swap/SecuritizeSwap.sol
contracts/token/DSToken.sol
contracts/token/IDSToken.sol
contracts/token/StandardToken.sol
contracts/token/TokenLibrary.sol
contracts/trust/IDSTrustService.sol
contracts/trust/TrustService.sol
contracts/utils/BaseDSContract.sol
contracts/utils/BulkBalanceChecker.sol
contracts/utils/CommonUtils.sol
contracts/utils/MultiSigWallet.sol
contracts/utils/TransactionRelayer.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The DSToken protocol implements a comprehensive regulated security token system designed to bridge traditional securities compliance with blockchain technology. At its core, the protocol consists of a rebasing ERC20 token `DSToken` that maintains regulatory compliance through an interconnected suite of smart contracts managing investor registration, transfer restrictions, and multi-jurisdictional regulatory requirements.

The architecture employs a modular service-oriented design where each component handles specific regulatory or operational functions. The `RegistryService` maintains KYC/AML investor records, tracking attributes like accreditation status and country of residence, while supporting multiple wallets per investor identity. The `ComplianceService` enforces transfer restrictions based on configurable rules including investor limits by jurisdiction, lockup periods for newly issued tokens, and minimum holding requirements. The system distinguishes between US accredited/non-accredited investors and EU qualified/retail investors, implementing different regulatory frameworks for each jurisdiction.

Token economics are managed through a rebasing mechanism where a multiplier adjusts token balances to reflect dividends or corporate actions without requiring explicit distributions. The protocol implements locking mechanisms

at both wallet and investor levels, including manual locks and automatic lockup periods for regulatory compliance. These locks use different accounting methods: manual locks track fixed token amounts while compliance lockups track share amounts that scale with rebasing events.

Token issuance flows through the `TokenIssuer` contract, which coordinates compliance validation, registry updates, and lock creation. Additional infrastructure includes a `MultiSigWallet`, a `TransactionRelayer` for pre-signed transactions, and a `SecuritizeSwap` contract enabling compliant token purchases using stablecoins with integrated KYC onboarding. The protocol also supports investor liquidation modes, cross-chain USDC bridging, and event logging for audit trails.

9 Attestation

Cyfrin attests that it performed a security review of the Securitize DSToken Rebasing code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-10-10-cyfrin-securitize-dstoken-rebasing-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=f2f7b600d694ec2977f057c9bb3d4ee838a165db9458f5d351b09f3211d84327
F=2025-10-10-cyfrin-securitize-dstoken-rebasing-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.